Assignment – 2.11

DNS Server Configuration

Q - 1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

- **Install and Set Up DNS Server Using BIND (Linux):**

Step 1: Update Linux

sudo apt update

Step 2: Install BIND9

sudo apt install bind9 -y

Step 3: Start DNS service

sudo systemctl start bind9

sudo systemctl enable bind9

Step 4: Create DNS zone

Open the configuration file:

sudo nano /etc/bind/named.conf.local

Add:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Step 5: Create zone file

```
sudo cp /etc/bind/db.local /etc/bind/db.musicmania.in
```

Step 6: Check configuration

```
sudo named-checkconf
sudo systemctl restart bind9
```

- **Result:**

Successfully installed the BIND DNS server and created the DNS zone musicmania.in on Linux.

Q - 2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

- **Add an A Record:**

Step 1: Open your DNS zone file and add:

www IN A 192.168.1.50

Step 2: Restart DNS server:

sudo systemctl restart bind9

Step 3: Check the DNS record:

nslookup www.musicmania.in

Expected Output:

Address: 192.168.1.50

- **Result:**

The A record is successfully created and maps www.musicmania.in to 192.168.1.50.

Q - 3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

- **Add a CNAME Record and Test:**

Step 1: Add a CNAME record in your DNS zone file:

play IN CNAME www.musicmania.in.

Step 2: Restart the DNS server:

sudo systemctl restart bind9

Step 3: Test using nslookup:

nslookup play.musicmania.in

Expected Output:

play.musicmania.in

canonical name = www.musicmania.in

Address: 192.168.1.50

- **Result:**

The CNAME record successfully maps play.musicmania.in to www.musicmania.in, which resolves to IP address 192.168.1.50.

Q - 4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Hint: Include at least three different troubleshooting commands or checks in your documentation.

- **DNS Troubleshooting Steps:**

1. **Check configuration:**
   Check DNS configuration files for mistakes.
2. **Check zone file:**
   sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
3. **Check configuration errors:**
   sudo named-checkconf
4. **Restart DNS server:**
   sudo systemctl restart bind9
5. **Check DNS service:**
   sudo systemctl status bind9
6. **Test DNS:**
   nslookup musicmania.in